

CS 436 and 536 Assignment 1

Description

1. Get the data: Take any number of years of Ford Ranger minimum and maximum price data from:
[Ford Ranger Models, Generations & Redesigns | Cars.com](https://www.cars.com/models/ford-ranger/generations-redesigns/)
2. Run two linear regression models (one for the minimum price and one for the maximum price). Use the following **error definition**, **e** and **cost function**, **J** when building and training both models:

$$e^{(i)} = h_{\theta}(x^{(i)}) - y^{(i)}$$
$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \left((e^{(i)})^2 + 0.1(e^{(i)})^4 \right)$$

Plot the loss curve for 100 iterations for each model.

3. You will notice that prices for 2012 to 2018 are missing from the URL. Predict what the Ranger prices would have been in those years if Ford had launched Ranger models then.
4. What are the final values of theta (weights) after training?
5. Plot the year-to-price curves with predicted values for the missing years (2012 to 2018).
6. Repeat the above process with feature scaling and a dynamic learning rate.
7. Plot the loss curves again. Did you see any improvements?
8. What would be the possible price range for the 2012 to 2018 Ranger models with feature scaling and a dynamic learning rate?
9. Plot the year-to-price curves with predicted values for the missing years (2012 to 2018) in this case as well.

Submission (Report PDF)

1. The total points will be reduced by 20% for each day after the due date.
2. **Submit a SINGLE PDF file including all code with comments, plots, answers and a summary.** See the points distribution below:
3. Code comments are required, including comments for the linear regression models, plotting, and prediction code. **[50 points]**
4. Plot all four models' loss curves and the year-to-price plot including the missing values. **[5 points each, 30 points total]**
5. Write a short summary paragraph. For example: What is the predicted price range of the next-generation Ranger (for 2026)? What are the final values of theta after training? What do you think of your model? How could you improve it? Compare performance. Would you buy a next-generation Ranger at that price? **[10 points]**
6. Write the formula for the gradient of the loss with respect to theta (weights) either by hand or print. **[10 points]**

Note

1. Include the entire notebook code in PDF format, merged into the report. Points will be deducted if this is not done.
2. You may also provide links to your notebook in the report.
3. Review and follow the Watson College Academic Honesty policies, which explain the consequences of academic dishonesty.
4. Do not copy code from or give code to others. If plagiarism is found, both students will receive zero points.
5. You can submit multiple times before the due date; only the last submission will be graded.
6. You may use library functions only for plotting (plot) and basic mathematical operations (e.g., exp, sum, mean). **No machine learning libraries are allowed.** Implement the linear regression models and training code yourself.